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Integrated conductive hydrogel soft actuators for remote photothermal actuation and multimodal self-sensing

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Conductive hydrogels are promising materials for advanced applications in artificial muscles, biomimetic soft robotics, and wearable electronics. However, the simultaneous realization of rapid reversible actuation, superior mechanical robustness, and high-resolution multimodal sensing remains a formidable challenge. Herein, we present a multifunctional hydrogel based on thermos-responsive poly(*N*-isopropylacrylamide) (PNIPAM), reinforced via acrylamide (AM) copolymerization and polyvinyl alcohol (PVA) network integration, which synergistically enhance mechanical strength and toughness. The incorporation of MXene nanosheets endows the hydrogel with stable, repeatable, and ultrasensitive piezoresistive sensing performance. Moreover, the hydrogel exhibits excellent photothermal actuation under near-infrared (NIR) irradiation, enabling remote, light-actuation deformation coupled with real-time self-sensing. To enrich its sensing modalities, a piezoelectric composite layer composed of poly(vinylidene fluoride-trifluoroethylene) and barium titanate [P(VDF-TrFE)/BTO] is integrated, allowing simultaneous detection of strain amplitude, movement direction, and velocity. As a proof of concept, a biomimetic octopus predation system was constructed, showcasing the potential of this integrated actuator-sensor platform for intelligent soft robotic systems.

1. Introduction

Bionics and hydrogels can be regarded as “twin technologies”, both aiming to develop smarter and more adaptive engineering systems.^{1,2} As technology continues to advance, human-machine interfaces are becoming increasingly complex, and traditional rigid structures can no longer meet the demands for flexible motion and rapid response. In contrast, hydrogels, with

New concepts

In recent years, self-sensing hydrogel actuators have attracted increasing attention due to their ability to monitor their own motion states in real time *via* electrical signals. However, existing systems remain largely limited to uniaxial sensing and face challenges in responding to complex multi-dimensional stimuli. This study proposes a bio-inspired hydrogel system capable of both photothermal actuation and multi-dimensional self-sensing, overcoming the limitation of uniaxial sensing in current self-sensing actuators. Its key innovation lies in addressing the lack of multi-dimensional information through a composite sensing mechanism combining piezoresistive (MXene nanosheets) and piezoelectric (P(VDF-TrFE)/BTO) effects, while simultaneously enhancing the mechanical strength of the hydrogel *via* a PNIPAM-AM/PVA double-network structure. Inspired by octopus predation behavior, this system integrates actuation, sensing, and control into a unified framework, enabling real-time feedback and autonomous responses in complex environments. It provides a new material paradigm with integrated “sensing-actuation-decision” capabilities for soft robotics.

their excellent deformability and scalability, demonstrate significant advantages in fields such as wearable electronics,^{3,4} soft robotics,^{5,6} electronic skin,⁷ and human-machine interaction.⁸ Nature has long served as a source of inspiration for the development of intelligent hydrogel materials.⁹ Animals produce mechanical motion through muscle contraction, while plants achieve morphological changes by regulating cellular water potential. For example, cephalopods (such as octopuses) exemplify a classic case of distributed sensing integrated with actuation: their flexible arms not only contain arrays of embedded mechanical sensors but also generate three-dimensional, non-linear motion through coordinated muscle fiber contractions, integrating proprioceptive information with external contact forces in real time and transmitting it to the central nervous system.^{10,11} Mimosa¹² and Venus flytrap,¹³ on the other hand, achieve rapid stimulus responses through directional water transport between parenchyma cells. The core inspiration from these biological systems lies in their embedded sensing-actuation integrated architecture, which realizes closed-loop

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environmental interaction through functional integration at the material level, offering new directions for the development of intelligent hydrogel actuators, namely, the need to create intelligent systems with stimulus-responsive and multi-physical sensing capabilities.

Currently, widely studied soft actuation materials include liquid crystal elastomers,^{14,15} electroactive polymers,¹⁶ shape memory polymers,¹⁷ and stimuli-responsive hydrogels.^{18,19} Among them, hydrogels are capable of achieving complex motion modes (such as rotation, rolling, crawling, and bending) under various stimuli, including pneumatic/hydraulic pressure,²⁰ electrical stimulation,²¹ light,²² humidity,²³ and magnetic fields.²⁴ Meanwhile, by integrating sensing units inside the hydrogel, external stimuli can be converted into various recordable signals such as resistance, current, capacitance, voltage, and optical color.^{25,26} This strategy endows the hydrogel with self-sensing and self-adaptive capabilities, enabling it to not only detect changes in the external environment but also monitor its own deformation state. For example, Tian *et al.*²⁷ used 3D extrusion printing to fabricate conductive hydrogel/dielectric elastomer composites. They introduced hygroscopic salt (LiCl) into the polyacrylamide hydrogel as a conductive and sensing material, enabling recognition of finger deformation with corresponding resistance signal outputs. Ma *et al.*²⁸ developed an artificial flower based on LM@A-MXene PNIPAM hydrogel that exhibited reversible opening-closing behavior under NIR light and could monitor its state in real time *via* resistance changes. Zhao *et al.*²⁹ employed an ice-templating method to prepare light/thermal-responsive PNIPAM/PANI composite hydrogels, integrating actuation and piezoresistive sensing in one material. This system can not only sense the external environment but also monitor its own deformation under stimulation, enabling self-monitoring of various delicate motions, including “contraction”, “bending”, “shape recognition”, and “object grasping and manipulation”. Additionally, Liu *et al.*³⁰ performed quantitative analysis of the actuation-sensing mechanism to achieve remote interactive control between soft and rigid robots, integrating precise control and monitoring through an Internet of Things (IoT) platform. Although hydrogel soft actuators have made significant progress in self-sensing functions, most existing systems are still limited to single-axis sensing and are difficult to cope with complex multi-dimensional stimuli. Although piezoresistive sensors can monitor the deformation amplitude and stress distribution in real time, they generally lack the ability to identify the force direction or capture the dynamic strain rate. These limitations are particularly prominent in the application of soft robots – where complex environmental interactions urgently require high-precision spatiotemporal force feedback. Therefore, there is an urgent need to develop advanced intelligent response materials that possess reversible deformation response, excellent mechanical properties, and multimodal sensing capabilities, in order to facilitate the development of highly adaptive soft robot platforms.

In this study, we developed a photothermal actuated hydrogel-based system with integrated multi-modal self-sensing capabilities, including piezoresistive strain sensing and piezoelectric dynamic force sensing. Specifically, a thermoresponsive

dual-network hydrogel was constructed using PNIPAM as the actuation component. By copolymerizing PNIPAM with acrylamide (AM) and incorporating a polyvinyl alcohol (PVA) reinforcing phase, the mechanical strength of the hydrogel was significantly improved. Subsequently, MXene nanosheets were introduced through a one-step process to enable both near-infrared (NIR)-triggered actuation and volumetric strain sensing. Additionally, a piezoelectric P(VDF-TrFE)/BaTiO₃ (PVDF/BTO) layer was integrated to detect directional dynamic forces during bending deformation. This integrated platform achieves autonomous motion through photothermal-induced deformation of the hydrogel while simultaneously monitoring movement in real time *via* the multi-modal sensing system, thereby enabling closed-loop control. Based on this platform, we designed and implemented a biomimetic octopus-inspired predation system that fully integrates actuation, sensing, and feedback mechanisms within a closed-loop framework.

2. Results and discussion

2.1 Structural design of the intelligent hydrogel system and the closed-loop system

A distinguishing feature of biological organisms, in contrast to artificial materials, lies in their intrinsic ability to dynamically perceive and regulate movement in response to environmental stimuli. Specifically, proprioceptors continuously capture localized motion (proprioception) and external stimuli (tactile perception), transmitting this information through reflex arcs or neuromuscular pathways to initiate timely responses *via* effectors. A representative example is the octopus (Fig. 1(a)), which leverages a distributed neural network embedded throughout its soft limbs to execute a broad spectrum of motions—from basic elongation and contraction to sophisticated tasks such as precision grasping, morphological transformation, and active hunting. Draw inspiration from the predatory behavior of octopus tentacles, this study presents an innovative bioinspired flexible intelligent system endowed with proprioceptive capabilities (Fig. 1(b)). This system achieves autonomous photothermal actuation with concurrent self-monitoring through integrated piezoresistive strain sensing and P(VDF-TrFE)-based piezoelectric dynamic force detection. By leveraging the complementary nature of piezoresistive and piezoelectric effects, the system enables enriched multimodal information acquisition, including force magnitude, direction, and frequency. Among them, the integrated hydrogel was prepared by a one-pot ultraviolet light polymerization method (Fig. S1). Due to the abundance of functional groups on the MXene surface, strong hydrogen bonding interactions form between MXene nanosheets and PNIPAM molecular chains, resulting in uniform dispersion and stable fixation within the PNIPAM hydrogel network. Benefiting from the superior photothermal conversion efficiency of MXene and the porous structure of the hydrogel, the resulting composite exhibits rapid and multidirectional actuation under NIR irradiation. It is worth noting that MXene, as an excellent conductive nanofiller, constructs a conductive network inside the hydrogel.

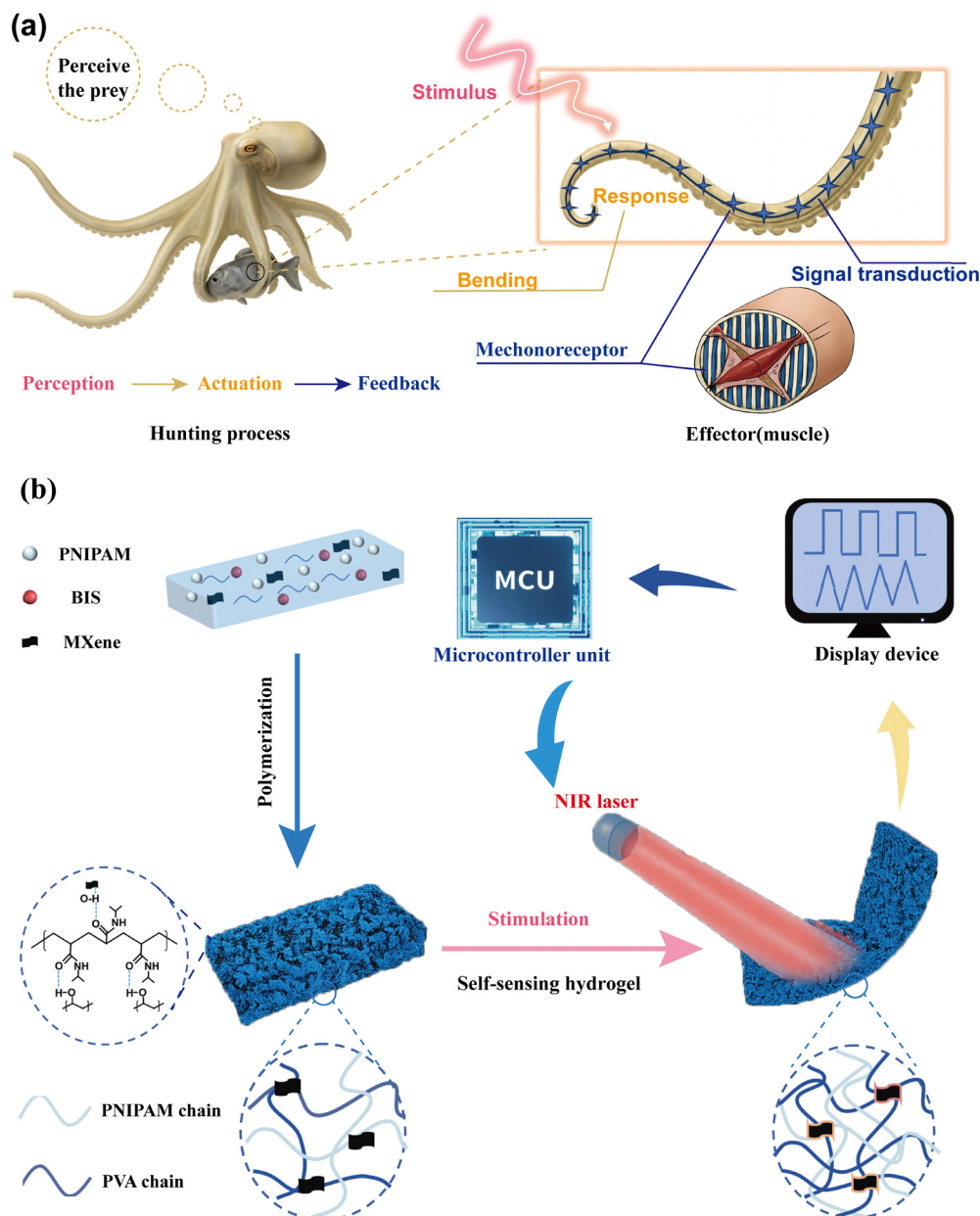


Fig. 1 (a) A schematic diagram of the octopus actuation process. (b) Schematic diagram of self-sensing hydrogel preparation and the closed-loop control system composed of it.

When the hydrogel deforms, the structure of this network changes accordingly, triggering a significant change in resistance. By monitoring this resistance change, the sensing of actuation can be realized. Moreover, the hydrogel's fast response and inherent sensing capability allow it to function simultaneously as both actuator and sensor. It can dynamically perceive external stimuli and autonomously transduce them into electrical signals in real time, thereby endowing the system with adaptive environmental awareness.

2.2 Microstructure and mechanical properties of hydrogels

The microstructure design of hydrogel critically governs their tunable functionality and performance. Hydrogels with

deliberately varied compositional ratios were synthesized to establish microstructure–property relationships. Scanning electron microscopy (SEM) of specimens reveals distinct microstructural features directly correlated with compositional differences. The SEM image of pure PNIPAM hydrogel is shown in Fig. 2(a), while the images of hydrogels containing AM and PVA are shown in Fig. S2. The hydrogels with AM/PVA addition exhibit a similar pore structure to pure PNIPAM hydrogels, both displaying open and uniform morphologies. However, the copolymerization reaction and the introduction of additional molecular chains enhance chain entanglement,³¹ which leads to a reduction in pore size. The SEM image of the hydrogel containing MXene is shown in Fig. 2(b). The incorporation of

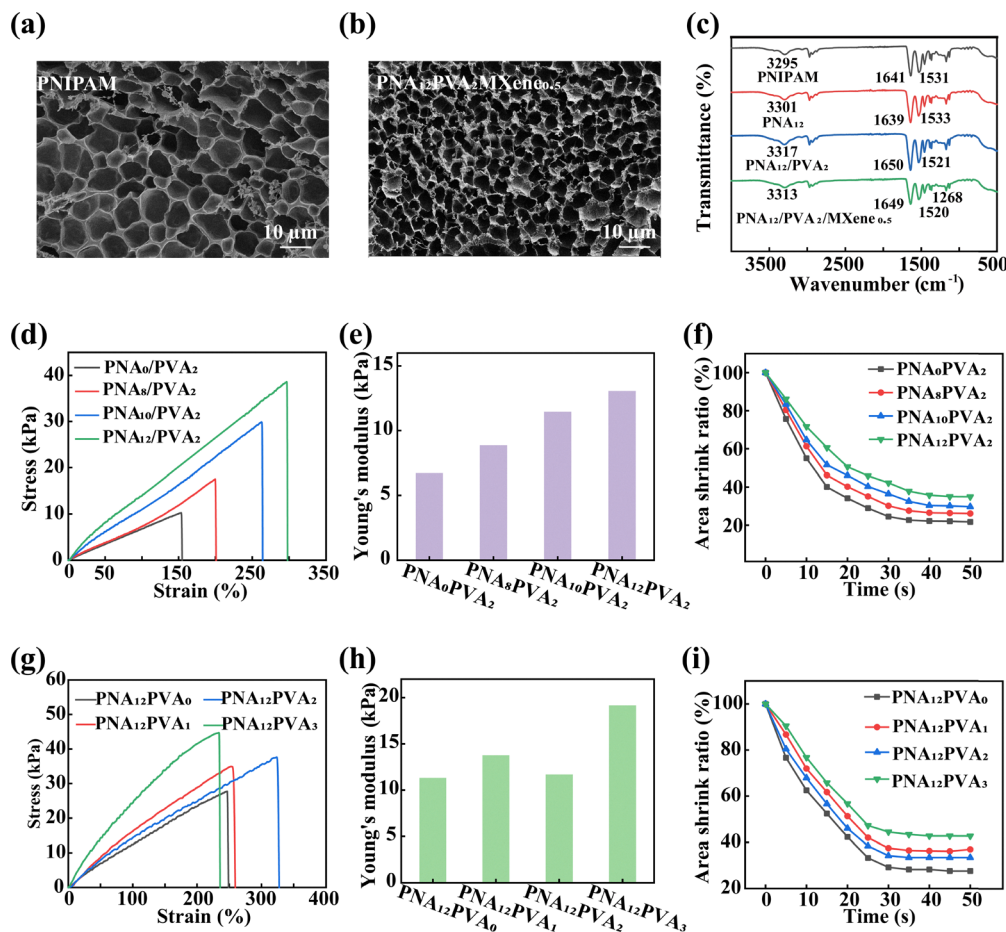


Fig. 2 (a) SEM image of PNIPAM. (b) SEM image of PNA₁₂/PVA₂/MXene_{0.5}. (c) FTIR images of different components hydrogel. (d) Stress–strain curves of hydrogels with different AM contents. (e) Young's modulus of hydrogels with different AM contents. (f) Response kinetics of hydrogels with different AM contents. (g) Stress–strain curves of hydrogels with different PVA concentrations. (h) Young's modulus of hydrogels with different PVA concentrations. (i) Shrink ratio of hydrogels with different PVA concentrations.

MXene introduces additional physical crosslinking points within the hydrogel network, resulting in a denser structure and a further reduction in pore size.³²

Fourier transform infrared (FTIR) spectroscopy analysis (Fig. 2(c)) confirms the molecular interactions between the components. The characteristic peaks of pure PNIPAM are located at 1641 cm⁻¹ (amide I, C=O stretching vibration) and 1531 cm⁻¹ (amide II, N–H bending vibration). The spectrum of PNA hydrogel closely resembles that of pure PNIPAM, with characteristic peaks remaining largely unchanged. This is likely because AM and PNIPAM share the same amide functional groups, leading to peak overlap without significant shifts. After the addition of PVA, the amide I peak shifts to 1650 cm⁻¹, the amide II peak shifts to 1521 cm⁻¹, and the N–H stretching vibration peak at 3301 cm⁻¹ shifts to 3317 cm⁻¹. This spectral shift arises from enhanced hydrogen bonding between the hydroxyl (–OH) groups of PVA and the carbonyl (C=O) groups of PNIPAM, which restricts intramolecular interaction between the –NH and C=O groups within PNIPAM.³³ Following MXene incorporation, the N–H stretching vibration peak shifts further, indicating hydrogen bonding interactions

between the MXene nanosheets and the hydrogel molecular chains.

To further explore the effect of different components on the mechanical properties of the hydrogel, the stress–strain curves were analyzed (Fig. 2(d)). The results show that as the AM content increases, the mechanical performance of the hydrogel significantly improves. The maximum stress of the PNA₁₂/PVA₂ hydrogel reaches 38.54 kPa, which is 3.76 times that of pure PNIPAM hydrogel, while its maximum strain increases to 296%, about twice that of pure PNIPAM hydrogel. The Young's modulus calculation (Fig. 2(e)) shows that the modulus of PNA₀/PVA₂ hydrogel is 6.65 kPa, while the modulus of PNA₁₂/PVA₂ hydrogel increases to 13 kPa, nearly doubling. This enhancement is attributed to the copolymerization effect of AM, which effectively improves the mechanical properties of the hydrogel. The introduction of AM densifies the network pore structure and increases the Young's modulus, while the combined effect of both AM and PVA suppresses the volumetric shrinkage of PNIPAM hydrogel. As shown in Fig. 2(f), the addition of AM slows down the rate of volume change in the hydrogel.

Further investigation of the effect of PVA content on the mechanical properties (Fig. 2(g) and (h)) reveals that adding

PVA enhances the hydrogel's elongation, tensile strength, and Young's modulus, indicating that PVA is an effective reinforcing component. However, when the PVA content is further increased, the hydrogel's elongation significantly decreases, while tensile strength and Young's modulus sharply increase. This phenomenon may be linked to the crystallization-induced reinforcement effect of PVA. While the increased crystallinity enhances the material's maximum stress-bearing capacity, it significantly reduces its strain capability, thus limiting the hydrogel's flexibility. In the water of 70 °C, the volume shrinkage rate of hydrogels with different PVA concentrations shows a trend of first decreasing and then increasing, which is consistent with that of Young's modulus. Among them, the

PNA₁₂/PVA₂ hydrogel has high elongation strain and a relatively low Young's modulus. During the shrinkage process, it exhibits the minimum network resistance, thus possessing greater deformability and enabling more significant volume shrinkage in a shorter time. As shown in Fig. 2(i), this hydrogel can shrink its volume to 34.89% of the original one within 50 seconds. The microstructure (SEM) and mechanical properties of hydrogels with different MXene contents are shown in Fig. S3 and S4. Increasing the MXene nanosheet content helps form a more uniform and dense three-dimensional network structure. And the hydrogel element mapping images prove and verify the successful introduction and uniform dispersion of MXene in the hydrogel matrix. However, excessive accumulation of

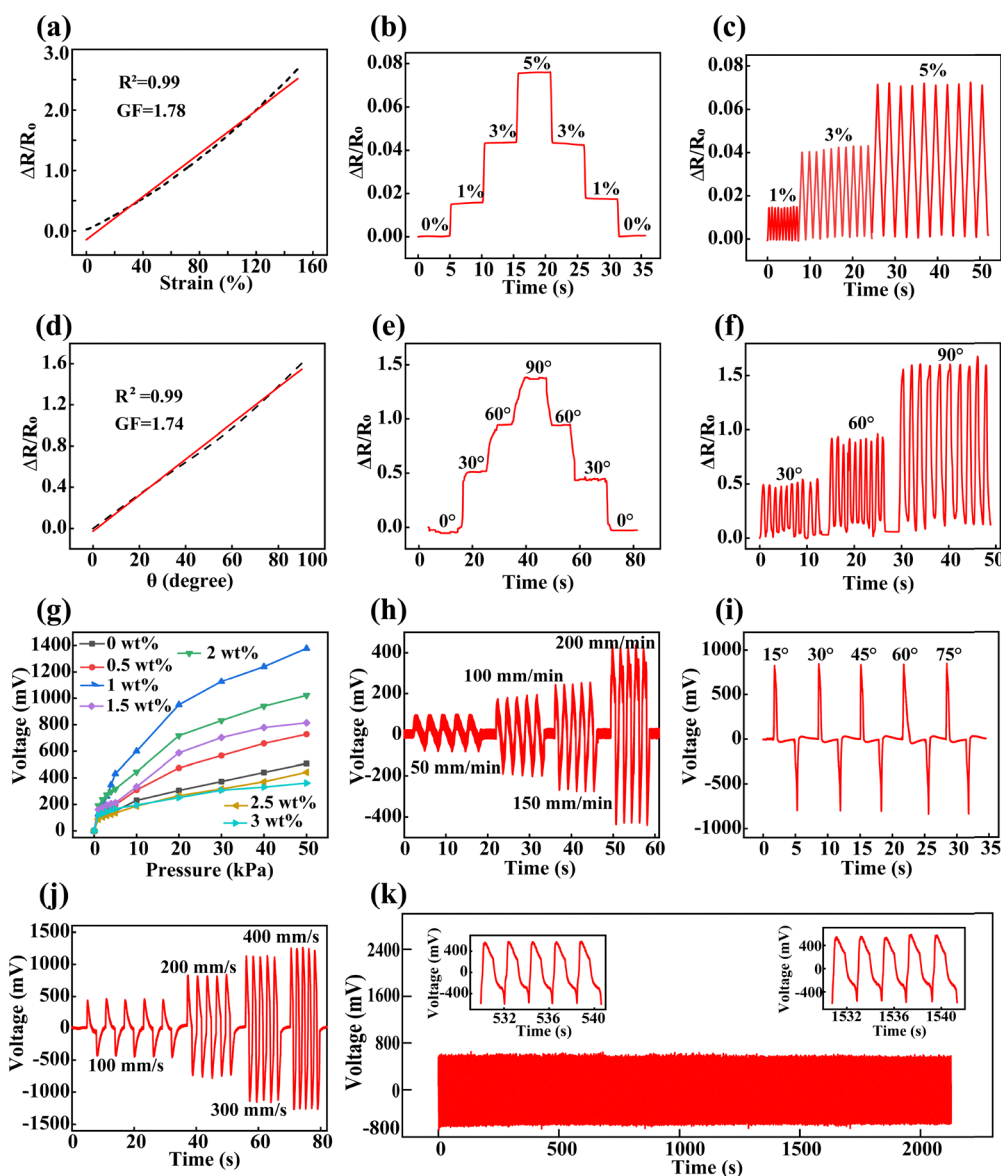


Fig. 3 (a) Relative resistance change ($\Delta R/R_0$) versus tensile strain. (b) Dynamic $\Delta R/R_0$ response during stepwise loading/unloading cycles. (c) Cyclic stability of $\Delta R/R_0$ under tensile strains of 1%, 3%, and 5%. (d) $\Delta R/R_0$ dependence on bending angle. (e) Variation of the hydrogel's $\Delta R/R_0$ with bending angle. (f) Repeatability of $\Delta R/R_0$ during multi-angle bending cycles. (g) Output voltage–pressure correlation for P(VDF-TrFE)/BTO films with varied BTO concentrations. (h) Output voltage response of hydrogel sensor under different loading rates. (i) Output voltage versus bending angle. (j) Output voltage dependence on bending rate. (k) Repeatability of the hydrogel sensor during 1000 bending processes.

MXene hinders polymer chain crosslinking, leading to defects in stress concentration, disruption of network continuity, and reduces effective crosslink density, which results in larger and more uneven pores. When the MXene concentration is 0.5 mg mL^{-1} , the hydrogel achieves optimal mechanical performance: the maximum tensile strain of 371.98%, the maximum stress of 48.35 kPa, and the Young's modulus of 13 kPa. Excellent mechanical properties are the basis for hydrogels to perform various actions. The subsequent research takes $\text{PNA}_{12}/\text{PVA}_2/\text{MXene}_{0.5}$ as the research object.

2.3 Multimodal sensing performance of integrated hydrogel-based systems

In addition to excellent mechanical properties, the hydrogel also demonstrates impressive multimodal sensing capabilities. By introducing MXene, the hydrogel acquires conductive properties, among which the electrical conductivity of the $\text{PNA}_{12}/\text{PVA}_2/\text{MXene}_{0.5}$ hydrogel could reach 0.06 S m^{-1} . To further explore the piezoresistive sensing properties of the hydrogel, we recorded the changes in relative resistance of hydrogels under mechanical stretching. As shown in Fig. 3(a), the hydrogel sensor exhibits excellent sensor signal linearity over a wide strain range ($\text{GF} = 1.78$). To characterize strain-dependent electrical response, strain steps from 0% to 5% were applied with intermediate holding and recovery phases. As shown in Fig. 3(b), the relative resistance change ($\Delta R/R_0$) increases proportionally with the applied strain, remains stable during the holding period, and quickly returns to the baseline upon release, demonstrating strong electrical stability. The sensor exhibits high reproducibility across repeated loading-unloading cycles at multiple strain amplitudes, as shown in Fig. 3(c) and Fig. S5. Consistent signal outputs across cycles confirm the hydrogel's good electrical stability and reproducibility at both high and low strains. Additionally, the sensor exhibits a fast response time (235 ms) and relaxation time (292 ms) (Fig. S6). To evaluate its motion-sensing performance, the hydrogel was induced to undergo bending deformation using a mechanical testing system. Within a range of 0° to 100° (Fig. 3(d)), the sensor's sensitivity coefficient GF is 1.74, demonstrating good linearity and sensitivity. Fig. 3(e) demonstrates discontinuous bending detection, where $\Delta R/R_0$ gradually increases with different bending angles. Upon returning to the original position, resistance gradually recovers to baseline, indicating dynamic reorganization of the conductive network. Repeated bending tests validate exceptional reproducibility and reliability across these angles (Fig. 3(f)), with consistent response retention over multiple cycles. Crucially, the integrated $\text{P(VDF-TrFE)}/\text{BTO}$ piezoelectric thin film enables simultaneous sensing of pressure magnitude, frequency, and bending directionality. Firstly, research was conducted on the properties of the prepared piezoelectric films. A moderate amount of BTO doping enhances the content of the ferroelectric β phase in the P(VDF-TrFE) matrix and improves its crystallinity, thereby enhancing the piezoelectric response (Fig. 3(g)). However, excessive BTO concentration disrupts the ordered arrangement of the P(VDF-TrFE) molecular chains, ultimately weakening its piezoelectric performance. The integration of the piezoelectric

thin film into the hydrogel enhances the output voltage in response to applied pressure (Fig. S7). Furthermore, under constant pressure, increasing the loading rate accelerates charge accumulation, leading to enhanced voltage output (Fig. 3(h)). The hydrogel also exhibits distinct frequency response characteristics. Additionally, the response time of the hydrogel based on piezoelectric sensing is 45 ms, which is favorable for capturing dynamic signals (Fig. S8). The piezoelectric sensor also demonstrates excellent reproducibility in 1000 long-cycle compression tests (Fig. S9). Furthermore, the embedded pre-polarized P(VDF-TrFE) layer generates piezoelectric voltage in response to mechanical bending. Bending tests of the thin film were conducted using a mechanical testing machine (Fig. S10), where the piezoelectric film outputs oppositely polarized voltage signals when bent in different directions (Fig. S11). The potential bending response mechanism is shown in Fig. S12. When a piezoelectric film is subjected to an external force and undergoes bending deformation, the piezoelectric effect causes polarized charges to accumulate on the electrode surface in an extremely short time. Specifically, during the bending process of piezoelectric films, the main external force they bear is the normal force, which directly prompts the piezoelectric film to produce the piezoelectric effect. Based on this correlation, the bending Angle of the piezoelectric film can be determined by detecting the magnitude of the output voltage. Fig. 3(i) demonstrates direction-dependent voltage generation: leftward bending induces a positive voltage peak that decays to baseline during relaxation, followed by a negative spike during snap-back recovery. Reversing the bending direction inverts this polarity sequence. While voltage amplitude remains independent of final bending angle at constant loading rates (Fig. 3(j)), it scales proportionally with increased strain rate due to accelerated piezoelectric charge accumulation. Extended cyclic bending tests confirm exceptional signal retention and durability with minimal drift (Fig. 3(k)).

2.4 Photothermal actuation and self-sensing characteristics of integrated sensor-actuator systems

Beyond its electrical functionality, the $\text{PNA}_{12}/\text{PVA}_2/\text{MXene}_{0.5}$ hydrogel exhibits exceptional photothermal actuation capabilities, significantly broadening its utility in intelligent systems. Under light irradiation, MXene acts as a broadband absorber, converting photons to thermal energy and inducing localized heating. Upon reaching the lower critical solution temperature (LCST), the hydrogel undergoes phase transition-driven dehydration, resulting in rapid shrinkage and directional bending. Fig. 4(a) shows that under unilateral infrared light irradiation, a temperature gradient is formed along the thickness direction of the hydrogel. This gradient directly causes the material to undergo uneven deformation, ultimately enabling the hydrogel to achieve directional bending. As shown in Fig. 4(b), the $\text{PNA}_{12}/\text{PVA}_2/\text{MXene}_{0.75}$ hydrogel can reach a high temperature of 128.7°C within 30 seconds under near-infrared light irradiation. The reason for this phenomenon lies in the fact that the hydrogel contains a large amount of water inside, and water has relatively poor thermal conductivity. As a result, the heat generated by the photothermal conversion of MXene cannot

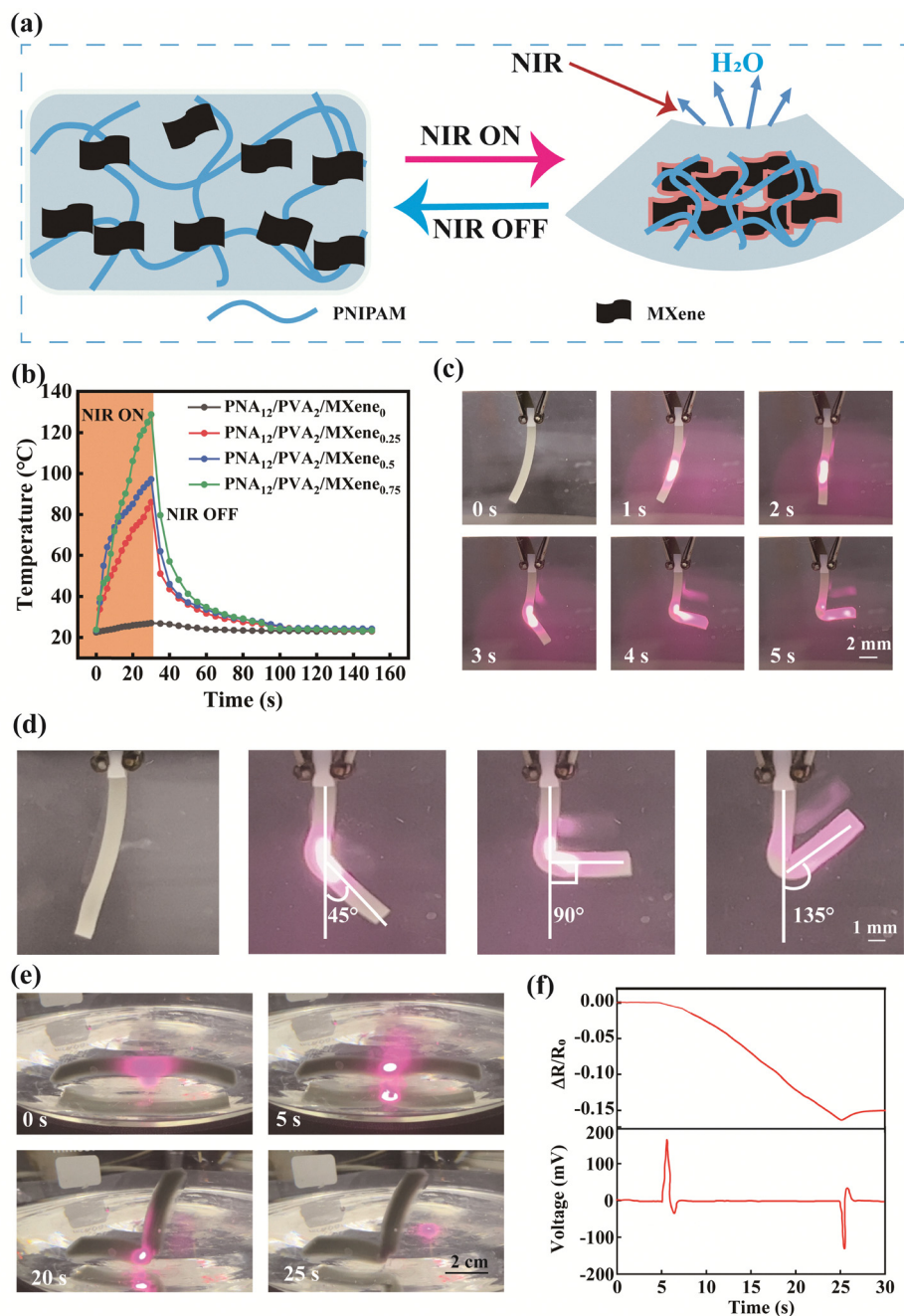


Fig. 4 (a) Dehydration-induced bending behavior of the integrated hydrogel. (b) Time-dependent temperature profile under NIR irradiation. (c) NIR-triggered bending kinetics of the integrated hydrogel in an aqueous environment. (d) Equilibrium bending angles of the integrated hydrogel under NIR irradiation in water. (e) Reversible bending response to cyclic NIR exposure in water. (f) Correlation between bending angle and real-time output voltage during self-actuation.

quickly diffuse throughout the entire system but accumulates in the irradiated area, eventually causing a sharp increase in local temperature. In contrast, the composite hydrogel without MXene maintains almost unchanged temperature, only increasing by 5 °C after 30 s of irradiation. After the infrared light is turned off, the temperature of the hydrogel quickly drops back to room temperature. The temperature of the PNA₁₂/PVA₂/MXene_{0.5} hydrogel under different light intensities is shown in Fig. S13. During the fixed 30 s irradiation, the hydrogel temperature rises with

increasing light intensity. Fig. 4(c) depicts the complete photo-actuation process of the hydrogel. When the hydrogel is exposed to NIR light, it bends rapidly to 90° within 5 seconds. Moreover, NIR light stimulation allows for more precise spatiotemporal control of the soft actuator. When NIR light is directed onto the hydrogel, it bends toward the direction of the incident light and precisely tracks the light source (Fig. 4(d) and Movie S1). Light irradiation induces local contraction of the hydrogel, which in turn promotes its bending. When it bends to a specific angle, the hydrogel itself

blocks the incident light, causing the driving process to stop and eventually achieving dynamic equilibrium. Fig. 4(e) and Movie S2 shows the entire process of the hydrogel in deionized water, actuation by NIR light. Upon exposure to NIR light, the hydrogel begins to bend and generates a positive voltage signal. During the bending process, the relative resistance change of the hydrogel gradually decreases. When the light source is removed, the hydrogel starts to recover and generates a negative voltage signal (Fig. 4(f)). The seamless integration of photothermal drive and multimodal self-sensing has established a multi-functional platform for the next generation of advanced flexible electronic devices, soft robots, and intelligent actuation systems.

2.5 Bio-octopus hunter: precision tracking with closed-loop control

Inspired by octopus tentacle predation dynamics, we constructed a biomimetic soft robotic arm with embedded proprioception. This system seamlessly integrates actuation and real-time state monitoring by harnessing the hydrogel's dual-mode sensing and photothermal actuation capabilities. As shown in Fig. 5(a), the hydrogel actuator is connected to a closed-loop control system *via* copper wires, allowing for real-time monitoring and analysis of feedback signals within the circuit. In this biomimetic framework, the hydrogel serves as

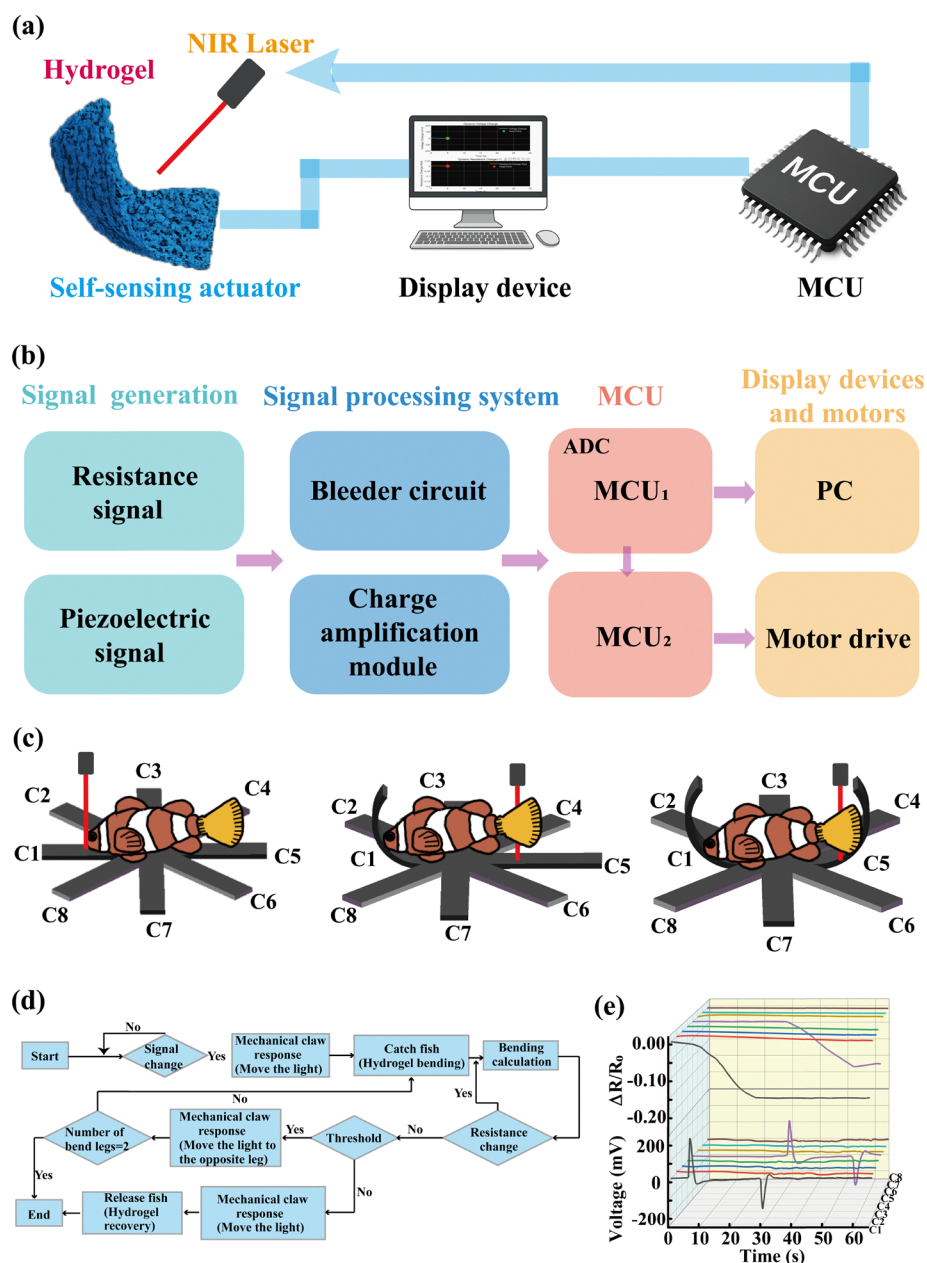


Fig. 5 (a) Schematic diagram of the bionic hunting system under NIR illumination. (b) Hardware logic block diagram of the bionic hunting system. (c) Schematic diagram of the fishing process of the bionic hunting series system in a water environment. (d) Flowchart of the bionic hunting process program. (e) Dual-mode perception signal diagram of the process of releasing animals for hunting.

both the receptor and effector, akin to the octopus. A micro-controller unit (MCU) functions as the central nervous system, analyzing these signals and coordinating adjustments, thereby assembling the modules into a biomimetic system capable of generating truly autonomous responses to external environmental stimuli. Fig. 5(b) illustrates the hardware block diagram of the hunting-release system. The main control chip is the STM32F103RCT6. Piezoresistive and piezoelectric signals are acquired through voltage divider circuits and charge amplifier circuits, respectively (Fig. S14 and S15). The CH340 module enables bidirectional communication between the acquisition system and the host computer. A secondary MCU controls the robotic arm, adjusting the position of the NIR light source upon receiving signals (Fig. S16). The specific process is detailed in Fig. 5(c) and (d). When prey makes contact with the biomimetic octopus, the robotic arm moves to direct the NIR light onto the corresponding octopus leg. The octopus leg begins to bend, gradually grasping the prey while simultaneously generating piezoelectric and piezoresistive signals from multiple channels (channel 1–8) (Fig. 5(e) and Movie S3). Specifically, after the prey triggers the movement of the robotic arm, the NIR light first illuminates the target octopus leg (C1), causing the octopus leg to gradually bend and grasp the prey. As shown in Fig. S17 and Movie S4, the experiment took a stationary toy fish as the target object. By regulating the irradiation of NIR, the directional bending of the hydrogel was achieved. During this process, channel 1 generates a positive voltage signal. Once the relative resistance change of channel 1 stabilizes and does not exceed a preset threshold, the robotic arm moves the NIR light to the opposite octopus leg (C5). At this point, channel 1 generates a reverse voltage signal. Under NIR light, the C5 leg begins to bend and participates in the grasping process. At this point, channel 5 generates a positive voltage signal, and its relative resistance starts to decrease. When the piezoresistive signal from channel 5 stabilizes and falls below the threshold, the robotic arm retracts the NIR light, and channel 5 generates a negative voltage signal, indicating the completion of the hunting process. Conversely, if the relative resistance change monitored by any channel exceeds the preset threshold during the hunting process, it indicates that the prey is too strong. In this case, the system triggers a release action and a defensive movement. It is worth noting that we set the threshold dividing line between “hunting” and “releasing” as the corresponding resistance value when the hydrogel is bent at 90°. When the hydrogel performs the hunting action and the bending Angle can be greater than 90°, it is determined that the hunting is successful. Conversely, it triggers the release process (Fig. S18). And we simulated and demonstrated the “hunting” process by attaching the hydrogel to the surface of a toy fish; when the hydrogel detaches from the surface of the toy fish, it corresponds to simulating the “releasing” process. This biomimetic system pioneers a material-based proprioception strategy for soft robotics. By integrating dual-mode piezoresistive–piezoelectric sensing with photothermal actuation, it achieves closed-loop manipulation autonomy–demonstrating precise object handling and showing transformative potential for

unstructured environments including deep-sea exploration and minimally invasive surgery.

3. Conclusion

This research work reports a strategy for a temperature-responsive double-network hydrogel based on PNIPAM. This hydrogel system integrates rapid reversible actuation, excellent mechanical properties, and multimodal sensing capabilities. Initially, a PNA/PVA hydrogel was fabricated by introducing AM copolymerization and a PVA reinforcement phase, which significantly enhanced its mechanical strength while preserving an open porous architecture. The hydrogel demonstrates outstanding photothermal actuation capabilities, shrinking to 34.89% of its original volume within 50 seconds in 70 °C water, underscoring its promise as an actuator. To further enable sensing functionality, MXene nanosheets were incorporated to form a PNA/PVA/MXene composite hydrogel. The resulting conductive network endowed the hydrogel with highly sensitive, stable, and repeatable piezoresistive sensing capabilities. Leveraging the high photothermal conversion efficiency of MXene and the hydrogel's porous structure, remote near-infrared (NIR) photothermal actuation was successfully achieved, integrating both actuation and sensing into a single platform. Finally, the introduction of P(VDF-TrFE)/BTO piezoelectric films enabled the system to detect not only strain magnitude but also its direction and velocity. The bionic octopus hunting system constructed based on this achieved a closed-loop operation of driving – sensing – feedback, demonstrating its application potential in soft robots and bionics.

4. Experimental section

4.1 Materials

N-Isopropylacrylamide ($C_6H_{11}NO$, 98%), *N,N'*-methylenebis(acrylamide) ($C_7H_{10}N_2O_2$, AR), Photoinitiator Darocur 1173 ($C_{10}H_{12}O_2$, AR), Barium titanate ($BaTiO_3$, AR) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd (China). Poly(vinyl alcohol) 1799 [$[C_2H_4O]_n$, hydrolysis > 99%), MXene ($Ti_3C_2T_x$, 5 mg mL⁻¹), Acrylamide (C_3H_5NO , AR), *N,N*-dimethylformamide (C_3H_7NO , AR) were supplied by Shanghai McLean Biochemical Co., Ltd (China). Polyvinylidene fluoride (vinylidene fluoride) ($[(C_2H_2F_2)_n(C_2HF_3)_m]$, 7:3) was obtained from the French chemical company Arkema.

4.2 Fabrication of P(VDF-TrFE)/BTO piezoelectric thin films

P(VDF-TrFE) powder (1.0 g) was dissolved in dimethylformamide (DMF, 8.9 g) under magnetic stirring at 60 °C. Barium titanate (BTO) nanoparticles (0.1 g) were dispersed into the solution, followed by 6 h stirring at 60 °C and 30 min ultrasonication. The homogeneous mixture was doctor-bladed into square-grooved molds, dried at 60 °C for 2 h, and demolded. Films were annealed at 120 °C for 2 h to enhance crystallinity. Electrical poling was performed in silicone oil *via* stepwise DC field application: 0.5 kV (30 min), 0.8 kV (60 min) and 1.0 kV

(30 min). Poled films were repeatedly rinsed with ethanol to remove silicone oil residues, yielding piezoelectrically active P(VDF-TrFE)/BTO thin films with optimized dipole alignment.

4.3 Preparation of the integrated intelligent hydrogel with actuation and dual-mode sensing

The composite hydrogel was prepared by a one-pot method. 1 g of NIPAM, 5 mg of BIS, 25 μL of 1173, and 120 mg of AM were added to 2.2 mL of deionized water and stirred evenly until completely dissolved. Subsequently, 1 mL of 10% PVA solution was added and stirred continuously until the precursor solution became clear and transparent. The MXene water dispersion system was ultrasonically dispersed for 30 minutes, and then added to the hydrogel precursor solution for 20 minutes of ultrasonic treatment. Immediately afterwards, the precursor solution was injected into a mold (1.5 mm thickness). Place the above-mentioned mold in an environment of 18 °C for ultraviolet light polymerization for 20 minutes, and maintain a constant temperature during the light exposure period. After the polymerization was completed, it was taken out and demolded to obtain the PNA/PVA/MXene composite hydrogel. Further, the PNA/PVA/MXene composite hydrogel was placed on a glass plate as the substrate, and the P(VDF-TrFE) piezoelectric film was laminated on the hydrogel. Then, place a 1 mm thick silicone pad on the original silicone pad and hold it in place with a glass slide. Inject the prepared AM hydrogel precursor into the mold that had been fabricated. Then place the above-mentioned mold in an environment of 18 °C and carry out ultraviolet light polymerization reaction for 20 min, maintaining a constant temperature during the light exposure period. After the polymerization was completed, it was taken out and demolded to obtain the driver-sensing integrated intelligent hydrogel. It was named $\text{PNA}_x/\text{PVA}_y/\text{MXene}_z$ hydrogel, where PNA stood for the copolymer hydrogel of NIPAM and AM. x , y , and z represented the content of each component. Among them, the AM precursor solution was prepared by completely dissolving 0.5 g of AM, 2 mg of BIS and 10 μL of 1173 in 2 mL of deionized water.

4.4 Characterization and testing methods

Hydrogel specimens were cryofractured in liquid nitrogen, freeze-dried at $-80\text{ }^{\circ}\text{C}$ for 24 h (FD-1C-50, Beijing Boyikang), and sputter-coated with gold prior to imaging. Microstructural morphology was characterized by scanning electron microscopy (SEM, Hitachi S-4800). The elemental distribution in the material was analyzed using energy dispersive X-ray spectroscopy (Horiba 7021-H). Chemical composition was analyzed *via* Fourier-transform infrared spectroscopy (FTIR, Thermo Scientific Nicolet iS20) with 4 cm^{-1} resolution across $500\text{--}4000\text{ cm}^{-1}$.

4.5 Test of mechanical properties

It was carried out on the HY-0230 stress-strain tester produced by Shanghai Hengyi Precision Instrument Co., Ltd. The sample preparation and testing process was as follows: the hydrogel was prepared into rectangular strips with dimensions of $40\text{ mm} \times 10\text{ mm} \times 1.5\text{ mm}$ through a mold. The hydrogel was fixed on

the mechanical testing machine using fixtures, and the program was set to conduct the test at a tensile rate of 50 mm min^{-1} . Record the test data and obtain the stress-strain curve and Young's modulus through calculation and analysis.

4.6 Response dynamics test

The prepared hydrogel was swollen to a balanced state in deionized water. The hydrogel strips were cut into hydrogel blocks of $10\text{ mm} \times 10\text{ mm} \times 1.5\text{ mm}$ using a surgical knife. The hydrogel blocks were placed in hot water at $70\text{ }^{\circ}\text{C}$. The process of their volume shrinking until it remained completely unchanged was recorded by a camera, and images at different times were taken. The area of the captured images was measured and calculated through the Image J software, and the relationship between the area of the hydrogel and time was obtained. To avoid errors, five samples were tested under each experimental condition to obtain statistically reliable values.

4.7 Test of sensing performance

The piezoresistive performance of the hydrogel was evaluated using a synchronous electromechanical test system. The universal testing machine (HY-0230, Shanghai Yiheng) was used to apply controllable uniaxial tensile force, while the corresponding resistance was continuously monitored using a Keithley 2400 digital source meter. To assess performance under bending deformation, the integrated hydrogel actuator was fixed onto a cardboard substrate. Different bending angles were then imposed using a mechanical tester, and the resulting piezoresistive response (resistance variation) was measured with the Keithley 2400 source meter. Simultaneously, the piezoelectric output voltage was continuously recorded using a charge amplifier.

4.8 Test of photothermal performance

Hydrogel specimens ($10\text{ mm} \times 10\text{ mm} \times 1.5\text{ mm}$) were mounted on a substrate. An NIR lamp was positioned 15 cm above the samples to ensure full illumination, and surface temperature distribution was recorded throughout irradiation using an infrared camera.

4.9 Test of optical drive deformation behavior

We fixed one end of the hydrogel with clips and freely suspended the other end in a cold deionized water environment. We irradiated the hydrogel with NIR light of a wavelength of 808 nm and a power of 1 W to cause it to bend. After bending to a certain extent, the light source was turned off, and the sample absorbed water and swollen to return to its initial state. The entire process was recorded by a camera. The images of the bending states at different time points were captured, and the bending angles of the hydrogel were measured using the Image J software.

Author contributions

Ping Guo: methodology, data curation, writing – original draft preparation; Jie Zhou: data curation, software, validation;

Chengnan Qian: data curation, software; Wenjie Cao: software, validation; Yang Yu: software, validation; Daoyou Guo: software, validation; Lin Cheng: formal analysis, investigation; Huaping Wu: investigation; Aiping Liu: supervision, conceptualization, methodology, writing – review & editing.

Conflicts of interest

The authors declare no conflict of interest.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supporting information: the hydrogel preparation process; SEM and EDS analyses; temperature and NIR effects on hydrogel programming; and actuation performance characterization. Movie S1: the photo-thermal actuation process of hydrogels underwater. Movie S2: the hydrogel is bent by near-infrared light in water while providing real-time feedback on changes in electrical signals. Movie S3: real-time signal changes in the hunting process of octopus legs. Movie S4: the grasping process of bionic hunting system. See DOI: <https://doi.org/10.1039/d5mh01647a>.

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